PHONETICS OF TAIWANESE MANDARIN

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1. Introduction

Mandarin Chinese, or just Mandarin, is a widely spoken dialect of Chinese. It was instated as the national language of china in the early 20th century and today around 80% of China's population speaks it (Weng, 2018) (Wong, 2019) (E'nuo & Yue, 2020). It is also spoken widely elsewhere; for instance it has been a national language of Taiwan since 1913, and is used in most schools ("Chapter 2: People and Language", 2011). For the purposes of this project, we will be focusing on the Taiwanese dialect of Mandarin. Unless otherwise stated, we mean the Taiwanese dialect of Mandarin when we discuss 'Mandarin'.

1.1. **The speaker.** The speaker participating in this project is Kaishin Lin. Kaishin was born in Taichung Taiwan in 2004 and lived there until age 18 when he came to the United States. Growing up in Taichung, Kaishin had a formal education in Taiwanese Mandarin and is literate. He also started learning English as a toddler alongside Mandarin. He currently attends Reed College, but returns to Taiwan for multiple months each year. At Reed he speaks Mandarin once or twice a week, talking to his family. In Taiwan he speaks it daily.

We met with Kaishin three times for the duration of this project. The first was routine, to collect background information on his life and his experience with Mandarin. The second was to have him produce a variety of words so we could get a sense for what sounds might exist in his speech. The final meeting was to record him. This was done on March 29, 2026. We recorded him using a blue yeti microphone attached to a laptop in the lab of linguistics at Reed College. The word list is in Appendix A and the corresponding sound files will be attached.

1.2. **Outline.** The remainder of the report will be devoted to analysis of the recordings of Kaishin. We will first discuss the various types of consonants he produced, followed by a discussion of the vowels, and finally some discussion on tones.

	Bilabial	Labio-dental	Alveolar	Retroflex	Palatal	Velar/ Glottal
Plosive	p p^h		t t^h			k k^h
Affricate			ts ts^h	ʈʂ $ʈʂ^h$	tɕ $tɕ^h$	
Nasal	m		n			ŋ
Fricative		f	s	ʂ	ɕ	h
Approximant/ Lateral	w		l	ɻ	j ɥ	w

TABLE 1. Observed consonant inventory from the samples of speech for Kaishin Lin. The labialized velar approximant /w/ is placed in both cells.

2. Consonants

We discuss the consonants observed in the voice recordings of the speaker. We start with stops, then move to affricates, nasals, fricatives, and finally the approximants and laterals. A table of the speaker's consonant inventory is included in Table 1. The words used to record sounds in this inventory are in Appendix A.

2.1. Stops.

As seen in Table 1, there were six stops observed in the speaker, these being bilabial stops, alveolar stops, and velar stops. There were no voiced stops, although each place of articulation had both aspirated and un-aspirated versions of the stop. Interestingly despite having retroflex affricates /ʈʂ, $ʈʂ^h$/, we did not observe a retroflex stop /ʈ/ or /$ʈ^h$/.

The voice onset times (VOT) for each stop ranged from 10-20 ms for unaspirated stops, and 80-120 ms for aspirated stops across the words recorded for the stops in Appendix A. Generally it seems as if the word was lengthened in the second production, including somewhat the production of the stop. This may be because the speaker was producing these words in different contexts than he might initially. Regardless it is unknown which production is more faithful to his speech, and it may be that such lengthening is a natural part of his speech, so the mean VOT values should be observed. These values are in the top section of Table 2.

These results are explained well by our understanding of stop-length and articulator mobility. The alveolar stops /t, t^h/ and bilabial stops /p, p^h/ were much quicker to produce compared to their velar counterparts.

The spectral peaks and formant transitions are generally as expected. The spectral peaks in /t^h/ were slightly lower than otherwise expected, around 4200 Hz, as seen in the left side of Figure 2. This may be because there is no need to distinguish /t^h/ from other stops or

English Gloss	Transcription	Consonant	VOT 1	VOT 2	mean VOT
Eight	/pa˥/	p	10.0	13.5	13.1
To climb	/pʰa˧˥/	pʰ	85.5	103.8	94.65
Issue	/ta˨˩/	t	11.2	11.9	11.6
She	/tʰa˥/	tʰ	89.6	90.5	90.1
Brother	/kɤ˥/	k	20.8	18.2	19.5
Subject	/kʰɤ˥/	kʰ	121.8	103.1	112.5
Property/Resources	/tsɨ˥/	ts	78.8	103.3	91.1
Blemish/Flaw	/tsʰɨ˥/	tsʰ	170.0	187.5	178.75
To know	/ʈʂɨ˥/	ʈʂ	94.9	91.9	93.4
To eat	/ʈʂʰɨ˥/	ʈʂʰ	200.7	184.5	192.6
Plus/To add/To give	/tɕa˥/	tɕ	92.3	73.5	82.9
To choke	/tɕʰa˥/	tɕʰ	166.4	129.7	148.1

Table 2. VOT for each stop (top) and affricate (bottom). The VOT for each recording of the word, as well as the mean VOT among the two is recorded. All times were measured in ms.

affricates. Indeed, the closest consonants were /tɕ/ or /tɕʰ/, but as we will discuss in the next subsection, these consonants are still significantly different.

2.2. **Affricates.** There were six affricates observed in the speaker. These were alveolar /ts, tsʰ/, retroflex /ʈʂ, ʈʂʰ/, and palatal /tɕ, tɕʰ/. Like with stops, there were no voiced affricates, and each place of articulation had both aspirated and unaspirated affricates.

As seen in Table 2, the VOT for each affricate ranged from 70-100 ms for unaspirated affricates, and 130-200 ms for aspirated affricates. Of note is the speaker's productions of the aspirated palatal affricate in the word 'To choke', /tɕʰa˥/. The VOT for the affricate here was much lower than any of the other VOT for aspirated affricates, and nearly 40 ms lower than the other production. This may have been an oddity with this particular production, as the length of the word was approximately 396 ms compared to the speakers first production of the word which was around 500 ms long. Without this production, the aspirated affricates ranged from 160 ms to 200 ms, around 100 ms than the unaspirated affricates.

Another interesting thing in the speaker's production of affricates is the quality of some of the affricates. For instance, as seen in Figure 1 with "Plus/To add/To give", /tɕa˥/, his first production had a very messy affricate, whereas his second was very clean. This is the most extreme instance of this happening, although it happened to a lesser extent with some other words, with some productions having double bursts and some appearing more clean.

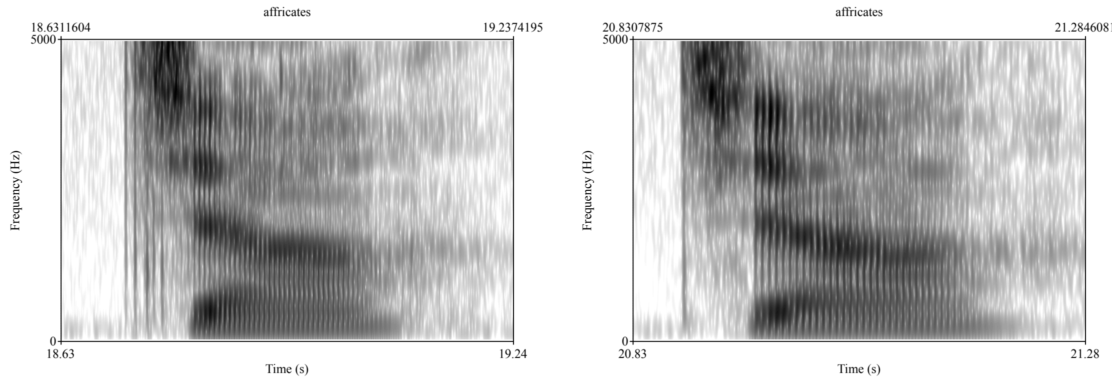

Figure 1. Spectograms of the speaker's two productions of "Plus/To add/To give", /tɕa/ (with a high tone). The unaspirated palatal affricate in the first production on the left is much messier than the second production on the right.

This variability is not unexpected in affricates, especially the palatal and retroflex affricates where the articulator has less mobility. Nonetheless we thought it worth observing.

Returning to the discussion on /t^{h}/ and /tɕ/ in subsection 2.1, we present a comparison of the two on the minimal pair /t^{h}a˥/ and /tɕa˥/ in Figure 2. As observed in Table 2, the VOT in /t^{h}/ is slightly longer than the VOT in /tɕ/. Most notably, the spectral peaks of /t^{h}/ and /tɕ/ differ significantly, and the formant transitions into the subsequent /a/ are quite different.

For instance, we see that the spectral peaks of /t^{h}/ are much broader after the initial burst and are similar to an /h/, as expected. Conversely, the /tɕ/ has its first spectral peak around 4300 Hz. Likewise in the formant transitions into /a/, we see a minimal change from /t^{h}/ but significant changes in /tɕ/. The formant transitions in /tɕ/ are predicted well by perturbation theory. Other cues, like burst quality, are not readily applicable to distinguish the two sounds because the burst quality of both affricates and aspirated stops will be quite messy.

A final note is on the inclusion of a retroflex affricate /ʈʂ/ instead of /tʃ/. We thought that it was more likely that the speaker had the former affricate based on the (lack of) formant transitions into the vowel /ɨ/ in "To know", /ʈʂɨ˥/. One of the speaker's productions of the word is in Figure 3. If the speaker was producing a /tʃ/ we would expect more extreme formant transitions into the backer vowel /ɨ/.

2.3. **Nasals.** There were only three observed nasals, in the same places of articulation as the stops, so /m,n,ŋ/. The literature suggests that /m/ can only occur in the onset of a

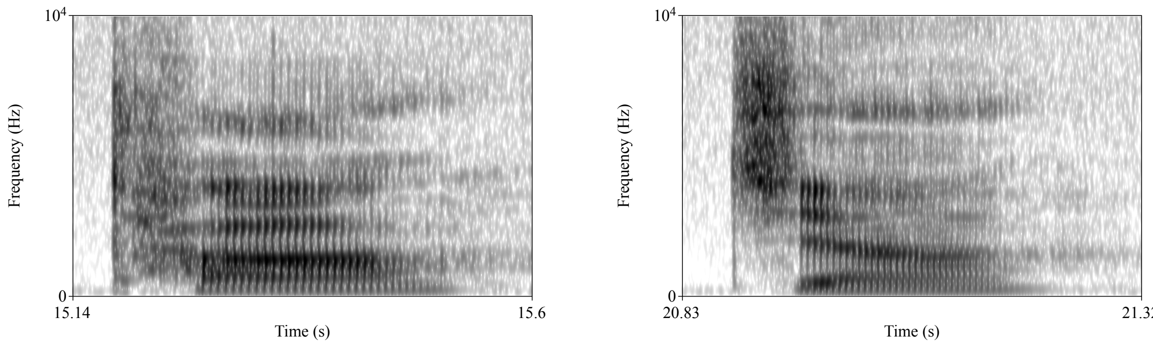

FIGURE 2. The speaker's first production of "She", /tʰa/ on the left and second production of "Plus/To add/To give", /tɕa/ on the right (both words have high tones).

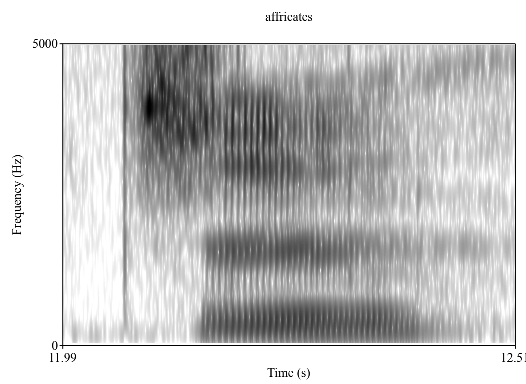

FIGURE 3. The speaker's production of "To know", /ʈʂɨ/ (with a high tone). Observe the relative lack of formant transitions into the vowel.

word, while /ŋ/ can only be in a coda (Li et al., 2017). This claim is in agreement with the words we recorded. Indeed /m/ only occurred at the beginning of words and /ŋ/ at the end. Although it may be that we neglected to find words which contradicted this. In contrast, it seems as if /n/ can occur in either position. We found instances in which /n/ was in either position, however we did not record words where /n/ was at the onset of a word.

Contrary to our understand of many nasals, the /m/ in the word for mother was quite loud. As shown in Figure 4 the higher formants were still have wider bandwidth and are slightly less clear, however they are definitely still present. There were not other recorded productions of /m/, so we cannot know if this is an oddity with the speaker's production in this case or a general trend. Regardless, the other two nasals behaved as we expected and generally had lower amplitudes.

2.4. **Fricatives.** We observed five fricatives in our recording, all voiceless. These being /f, s, ʂ, ɕ, h/. The literature suggested the existence of a voiced palatal /ʑ/, however we did

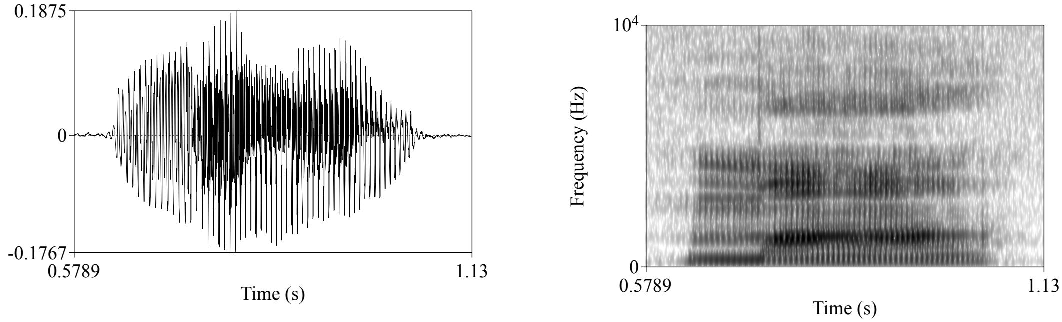

FIGURE 4. The waveform (left) and spectrogram (right) for "mother", /ma/ (with a high tone). Observe the intensity of the initial nasal.

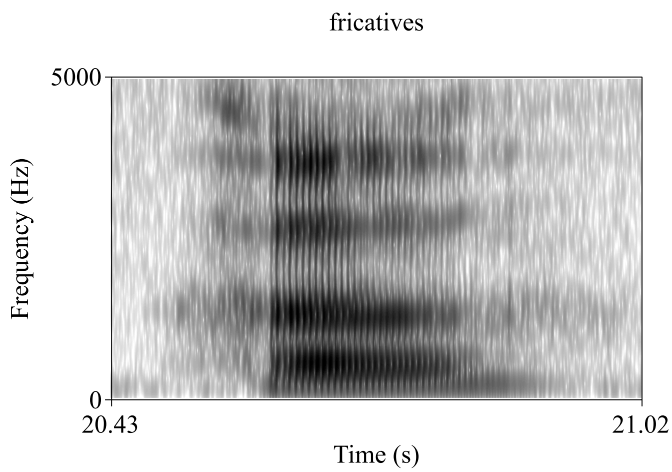

FIGURE 5. Speaker's pronunciation of "To drink", [hɤ] (with a high tone). Note the realization of the first fricative as a [h] instead of an expected [x].

not observe this sound (Kong et al., 2023). Indeed, the speaker had alternative translations for every word we found with an expected /ʑ/. Other sources suggested that there could be a /ʐ/ (Wan & Jaeger, 2003), and others noted that this could be realized as an /ɻ/ (Lin, 2025). We observed the latter in our speaker as in his translation of "To enter (a door)", which he pronounced as /ɻu˥˩/ instead of the possible alternative /ʐu˥˩/.

The literature also suggested the existence of a velar fricative /x/ (Kong et al., 2023) (Chen & Guo, 2022) (Li et al., 2017), but we observed that this was realized as a glottal fricative /h/ as suggested by Lin (2025). Indeed, as seen in Figure 5, the spectrogram of the speaker's production of "To drink" (expected [xɤ˥], observed [hɤ˥]) lacks the velar pinch characteristic of a velar fricative and instead the spectral peaks mimic the subsequent vowel, as expected of a glottal fricative.

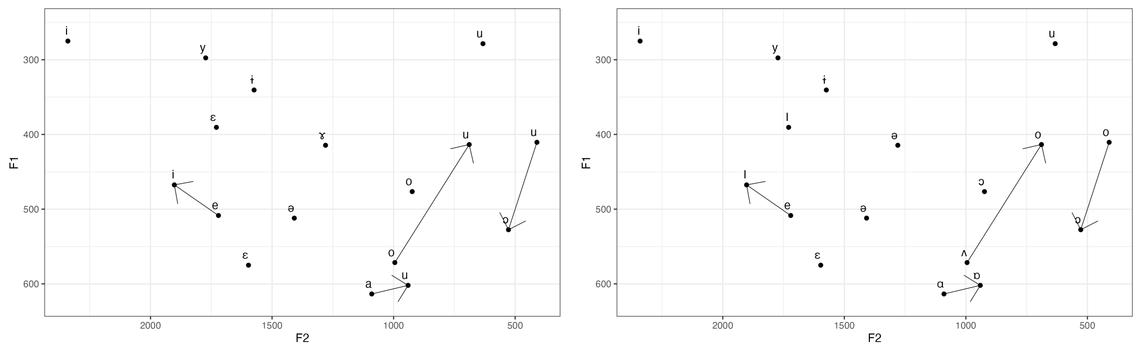

FIGURE 6. Vowel chart of the speaker. Measurements of F1 and F2 were taken at a central portion of each vowel for each production and averaged. The left figure shows the F1 and F2 values with the predicted vowels and transitions between diphthongs. The right figure has an annotation from Table 3.

English Gloss	IPA transcription	Predicted Vowel	Observed Vowel
Deer	/lu˥˩/	u	u
Is/Are/Yes	/ʂɨ˥˩/	ɨ	ɨ
Power/Ability	/li˥˩/	i	i
Green	/ly˥˩/	y	y̟
To extend/To apply (for a job)	/ʂən˥/	ə	ɜ
Subject	/kʰɤ˥/	ɤ	ə̞˞
Common/Together/Share	/koŋ˥˩/	o	ɔ̃
Side/Edge	/pjɛn˥/	ɛ	ɛ̃
Bill/List/Only	/tan˥/	a	a
(A) quilt	/pei˥˩/	ei	e̠ɪ
(of the sun) To shine on/To bask in	/ʂaɛ˥˩/	aɛ	aɛ
Empress	/hou˥˩/	ou	ʌo
To test	/kʰau˨˩/	au	ɑɒ
Fire	/huɔ˨˩/	uɔ	o̠ɔ̠

TABLE 3. Annotations in the vowels produced by the speaker.

2.5. **Approximants and Laterals.** We observed four approximants and one lateral. These were /w, ɻ, j, ɥ/, and /l/. These types of sounds were generally predicted by the literature. There are some slight deviations, for example the speaker had a /ɻ/ instead of a /ɹ/ or /ʐ/. These are minor however. As discussed in subsection 2.4, the expression of /ɻ/ as an approximant instead of a voiced fricative /ʐ/ is not unexpected of a speaker from Taiwan. We found that the most notable feature of the speaker's liquid's were the realization of /ɻ/ instead of /ʐ/ which we have already discussed, and so we move on to vowels.

3. Vowels and Tones

We discuss the vowels and tones observed in the speaker. We discuss different ways the vowels may have been realized in the speaker's production of the vowels. Finally we discuss the different tones.

3.1. **Vowels.** We present the F1 and F2 measurements for all the vowels and diphthongs. The vowel chart of the speaker is shown in the left side of Figure 6. He had monothongs /u, ɨ, i, y, ə, ɤ, o, ɛ, a/ and diphthongs /ei, aɛ, ou, au, uɔ/. Such monothongs and diphthongs are not distant from those reported in the literature (Wan & Jaeger, 2003) (Lin, 2025) (Li et al., 2017). We propose that these were realized as vowels as in Table 3, resulting in the plot on the right side of Figure 6.

It is difficult to be confident about some of these claims however. Many of the vowels were measured in places preceding a nasal, possibly biasing the measurement. The measurement of the schwa, for instance, was directly in front of a nasal. For at least the schwa, it may be hard to get un-nasalized productions as some sources have observed that a schwa can only be realized before a nasal /n/ or /ŋ/ (Wan & Jaeger, 2003). Of course it may be though that these vowels only exist in context with a nasal in which case considering their production in other contexts would not be reasonable.

Regardless, we claim that the /ɤ/ vowels were realized like a schwa. There were twelve different productions of this vowel throughout the recording; Figure 7 shows all the individual measurement superimposed on the vowel chart on the right of Figure 6. There are some possibly outlying points, however the hypothesis that /ɤ/ is being realized as a schwa seems well founded. It is still higher than the known schwa, however that may be accounted for by the nasalization in the production of schwa. It may also be that with a schwa only occurring before a nasal, distinguishing a /ɤ/ produced as a schwa from a real schwa may not be difficult for speakers.

Some other discrepancies from expectations, like the production /o/ as /o̞/ may also be attributed to nasalization, although there are not un-nazalized productions of this vowel recorded, so there is no way to verify such a claim. The initial transcription of the diphthong in "To test" as /au/ was incorrect however. Upon listening closely to the speaker's recording,

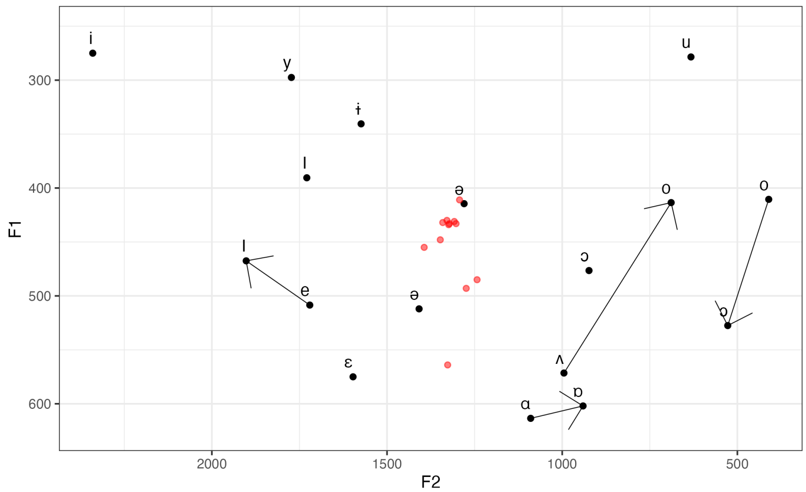

FIGURE 7. The annotated vowel chart from Figure 6 as well as all productions of the /ɤ/ vowel in red.

we transcribe this as instead /ɑɒ/. The transcription as /au/ was likely due to bias from our experience with the pre-existing diphthong /au/ in english.

One final note was on the productions of /uɔ/ (realized [o̯ɔ̯]). When producing these, it seems as if the speaker had very low F1 and F2 values such that the two were nearly indistinguishable. In Figure 8 we see a spectrogram of the speaker's production of the vowel. It appears as if the first two formants, predicted to be both very low, are nearly indistinguishable. It may be possible to separate them by hand, however that would be quite error prone, and we could not get Praat's formant tracer to distinguish between the line. As such we set the F1 and F2 values as equal as the only reasonable way to measure the two formants. This is certainly not ideal, but given the circumstances we thought it the best option.

3.2. **Tones.** We observed four tones in the speaker. These were /˥, ˥˩, ˧˥, ˨˩/. Some literature suggests that there may be a fifth tone, however we did not observe such a tone in a contrastive way in our recordings (Kong et al., 2023).

Other literature suggests that there may be a dipping tone /˨˩˧/ instead of a low-falling tone /˨˩/ (Li et al., 2017). We observed that latter initially and transcribed it as such. As we will discuss, a transcription of this tone as /˨˩˧/ may not be unreasonable, however transcribing it as /˨˩/ is reasonable given the literature and so we chose to continue with transcription as the /˨˩/ tone (Wan & Jaeger, 2003).

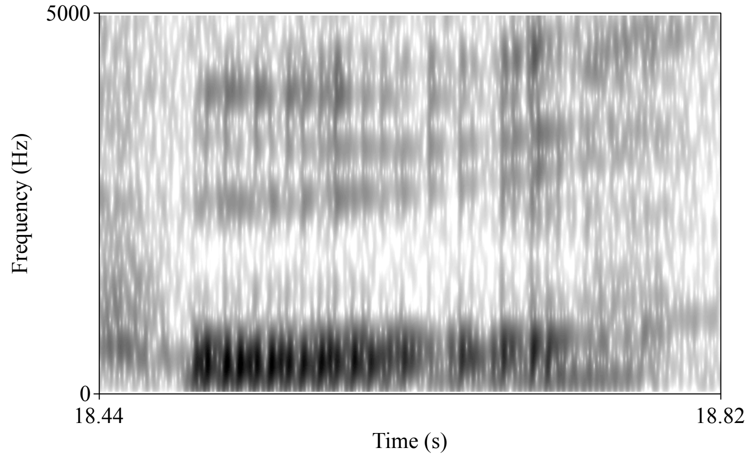

FIGURE 8. The speaker's production of the /uɔ/ diphthong. The first distinguishable formant is centered around 480 Hz, with the second around 2500 Hz.

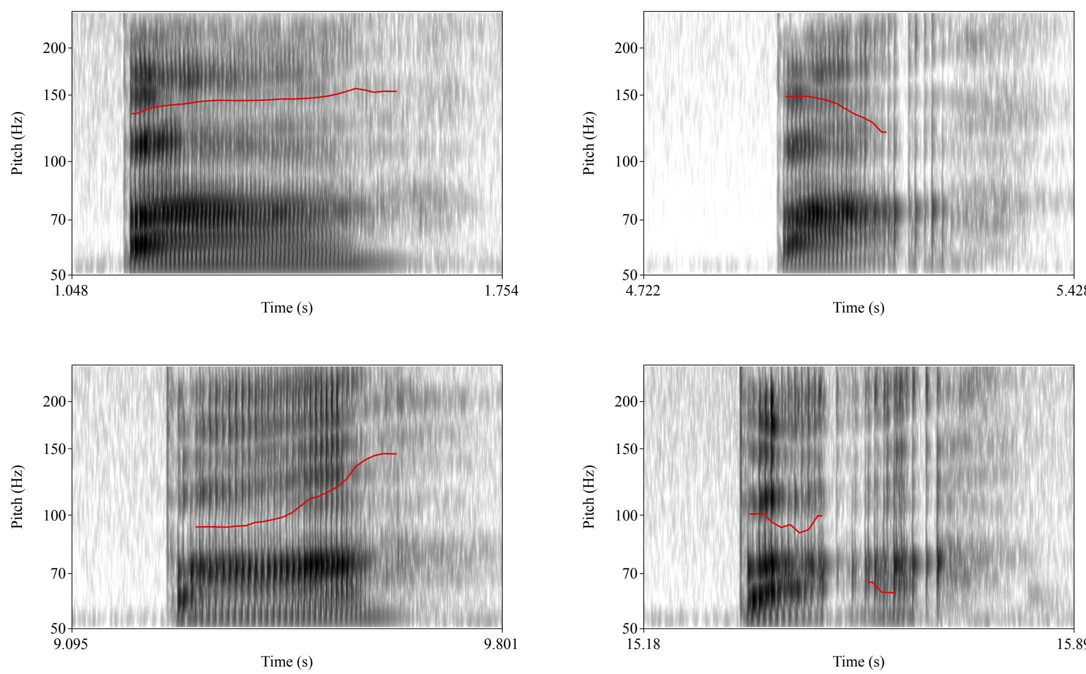

FIGURE 9. Spectrograms of one production of each of the four tones. The top left is the high tone, the top right is the high falling tone, the bottom left is the low rising tone, and the bottom right is the low falling tone.

The speaker's production with the minimal pairs of /pa/ (/pa˥/, /pa˥˩/, /pa˩˥/, /pa˨˩/) as well as pitch contours are included below in Figure 9. The associated values measured as the starting and ending pitches are in Table 4. We note that the measurement for "Eight", /pa˨˩/ was different then simply measuring pitch at the start and end of the production as

English Gloss	IPA transcription	Starting pitch (Hz)	Ending pitch (Hz)	Time of production (ms)
Eight	/pa˥/	133	154	491
Dad	/pa˥˩/	150	123	286
Pull	/pa˩˥/	92	146	399
Put	/pa˨˩/	95	64	338

Table 4. Minimal pairs for recording pitch with the starting pitch and ending pitch as well as word length. We used the second production of "Put" as shown in Figure 9 and measured the starting pitch as the average pitch in the first section, and the ending pitch as the average in the second section. Word length was measured as the time from the plosive until voicing stopped.

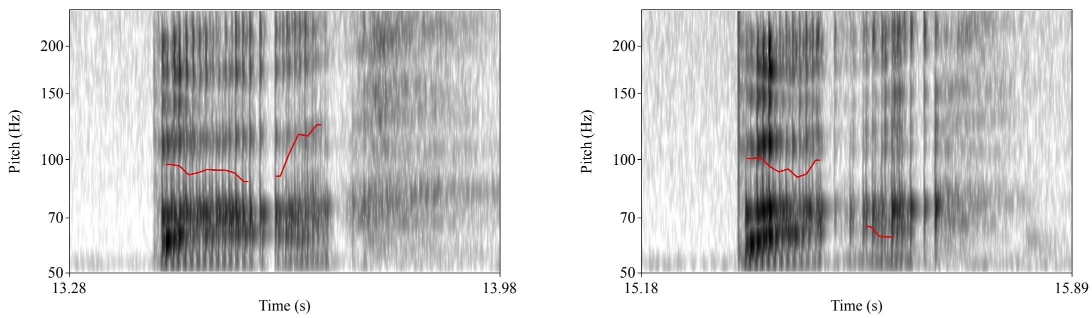

Figure 10. A comparison of the two productions of the low falling tone. The left image is the first production and the right image is the second production.

there were two disconnected segments. Instead we measured the starting pitch as the average pitch in the first section, and the ending pitch as the average in the second pitch. This is not necessarily accurate to pitch as something which changes throughout the word, but given the disconnect between the two parts of the pitch, we thought it was appropriate.

The pitch values suggest that the the high tone is realized close to a mid-high rising tone like [˧˥], and the falling tone is realized like [˥˧]. The rising tone still seems to be realized as a rising tone, but given the outlying pitch value in the ending pitch of "Put", we suggest that possibly the rising tone is realized as a [˩˥], so that the low falling tone is realized as [˨˩]. The specific transcription of these tones are consistent with those presented in some articles of the literature (Wan & Jaeger, 2003).

The length of the word may also affect how we perceive pitches. Although the high pitch /˥/ ([˥˧]) and falling /˥˩/ ([˧˥]) have very similar changes in pitch, the production of /pa˥/ is almost twice as long as /pa˥˩/. The lengthening of the word may de-emphasize the change in pitch, making a listener perceive the pitches in the high tone as less pronounced than in the falling tone. This would explain why the speaker produces a high tone as a slightly falling tone without it being misconstrued as another tone.

As mentioned briefly above, the speaker's production of the low falling tone /˨˩/ was not quite as expected. In his second production, his pitch has two disconnected segments. Further, as shown in Figure 10, one of the production had the second pitch raise, as we might expect of a dipping tone like /˨˩˦/. As they were produced within the same word, we suggest that the distinction between these tones is not contrastive in the speaker. Alternatively, there may be other phonetic cues to signal the meaning of the spoken word, or the speaker may simply have merged the two tones /˨˩/ and /˨˩˦/, possibly explaining why a fifth tone was absent.

4. Conclusions

We investigated the speech of a native speaker of the Taiwanese dialect of Mandarin Chinese. We found many broader trends with his consonants to be congruent with parts the literature we investigated, specifically Lin (2025). The notable differences are the production as a voiced retroflex fricative /ʐ/ as a retroflex approximant /ɻ/ and a louder nasal /m/. His production of vowels differed a little bit more, with some vowels like /ɤ/ being produced in a more fronted position, or changes in rounding.

A more expansive set of words created with care taken to nasalization may help understand some of the differences with vowels like /ə/ or /o/. Some vowels, like the fronted /ɤ/ are likely specific to our speaker though. Finally, we observed that the tones, while lacking a fifth tone which some sources suggested, seem to be congruent with the observations of Wan and Jaeger (2003).

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A. Word List

These are the recorded words. There are two repeated words in this list, these being "Brother" and "To drink". This was accidental but used in the report. Repeated words have been marked. Sources Wan and Jaeger (2003), Li et al. (2017), Chen and Guo (2022), Kong et al. (2023), Johnson (2020), and Lin (2025) were used for compiling this list.

Stops:

1. /p/ /pa˥/ Eight
2. /pʰ/ /pʰa˧˥/ To climb
3. /t/ /ta˨˩˦/ Issue
4. /tʰ/ /tʰa˥/ She
5. /k/ /kɤ˥/ Brother
6. /kʰ/ /kʰɤ˥/ Subject

Affricates:

7. /ts/ /tsɨ˥/ Property/Resources
8. /tsʰ//tsʰɨ˥/ Blemish/Flaw
9. /tʂ/ /tʂɨ˥/ To know
10. /tʂʰ//tʂʰɨ˥/ To eat
11. /tɕ/ /tɕa˥/ Plus/To add/To give
12. /tɕʰ//tɕʰa˥/ To choke

Nasals:

13. /m/ /ma˥/ Mother
14. /n/ /tan˥/ Bill/List/Only
15. /ŋ/ /toŋ˥/ Winter

Fricatives:

16. /f/ /fa˥/ Issue
17. /s/ /sɨ˥/ Silk
18. /ʂ/ /ʂɨ˥/ Teacher
19. /ɕ/ /ɕa˥/ Blind
20. /h/ /hɤ˥/ To drink

Approximants/Laterals:

21. /w/ /wa˨˩˦/ Tile
22. /ɻ/ /ɻu˥˩/ To enter (a door)
23. /j/ /jou˨˩˦/ To have/There is/To be
24. /ɥ/ /ɥe˥/ To make an appointment
25. /l/ /la˥/ To pull/draw/chat

Monothongs:

26. /u/ /lu˥˩/ Deer
27. /ɨ/ /ʂɨ˥˩/ Is/Are/Yes
28. /i/ /li˥˩/ Power/Ability
29. /y/ /ly˥˩/ Green
30. /ə/ /ʂən˥/ To extend/To apply (for a job)
31. /ɤ/ /kʰɤ˥/ Subject (Repeated)
32. /o/ /koŋ˥˩/ Common/Together/Share
33. /ɛ/ /pjɛn˥/ Side/Edge
34. /a/ /tan˥/ Bill/List/Only

Diphthongs:

35. /ei/ /pei˥˩/ (A) quilt
36. /aɛ/ /ʂaɛ˥˩/ (of the sun) To shine on/To bask in
37. /ou/ /hou˥˩/ Empress
38 /au/ /kʰau˨˩˦/ To test
39. /uɔ/ /huɔ˨˩˦/ Fire

Tones:

40. /˥/ /pa˥/ Eight
41. /˥˩/ /pa˥˩/ Dad
42. /˧˥/ /pa˧˥/ Pull
43. /˨˩˦/ /pa˨˩˦/ Put

Extra Words:

44. /kʰɤ˥˩/ Lesson
45. /hɤ˥/ To drink (Repeated)